

NYU, Tandon School of Engineering

Bridge to CS

Homework #2

Submission Instructions

1. For this assignment, you should turn in **4 .cpp files**, one for each coding question (Questions 1–4).
2. Name your files:
 - hw2_q1.cpp
 - hw2_q2.cpp
 - hw2_q3.cpp
 - hw2_q4.cpp
3. You should submit your homework in the Gradescope system.
4. For the coding questions, pay special attention to the style of your code: indent correctly, choose meaningful variable names, define constants where needed, and choose the most appropriate control flow statements.
5. **Tip:** For this homework, it is often a good idea to **avoid writing additional helper functions** and to **avoid using loops (e.g., for-loops)** unless the problem explicitly calls for them.

Question 1

Write a program that computes how much a customer has to pay after purchasing two items. The price is calculated according to the following rules:

- Buy one get one half off promotion: the lower price item is half price.
- If the customer is a club card member, additional 10% off.
- Tax is added.

Inputs to the program include:

- Two items' prices
- Have club card or not (User enters Y or y for “yes”; N or n for “no”)
- Tax rate (User enters the percentage as a number; for example they enter 8.25 if the tax rate is 8.25%)

Program displays:

- Base price — the price before the discounts and taxes
- Price after discounts — the price after the buy one get one half off promotion and the member's discount, if applicable
- Total price — the amount of money the customer has to pay (after tax)

Your program should interact with the user **exactly** as it shows in the following example:

```
Enter price of first item: 10
Enter price of second item: 20
Does customer have a club card? (Y/N): y
Enter tax rate, e.g. 5.5 for 5.5% tax: 8.25
Base price: 30.0
Price after discounts: 22.5
Total price: 24.35625
```

Question 2

Write a program that:

- Asks the user for their name.
- Asks the user to input their graduation year.
- Asks the user to input the current year.

Assume the student is in a four-year undergraduate program. Display the current status the student is in. Possible status include: not in college yet, freshman, sophomore, junior, senior, graduated.

Note: If graduation year equals to current year, status is **Graduated**; if graduation year is four years after current year, status is **Freshman**, etc.

Your program should interact with the user **exactly** as it shows in the following example:

```
Please enter your name: Jessica
Please enter your graduation year: 2019
Please enter current year: 2015
Jessica, you are a Freshman
```

Question 3

Write a program that does the following:

- Ask user to input three Real numbers a , b and c . They represent the parameters of a quadratic equation $ax^2 + bx + c = 0$.
- Classify to one of the following:
 - Infinite number of solutions (for example, $0x^2 + 0x + 0 = 0$ has infinite number of solutions)
 - No solution (for example, $0x^2 + 0x + 4 = 0$ has no solution)
 - No real solution (for example, $x^2 + 4 = 0$ has no real solutions)
 - One real solution
 - Two real solutions

- In cases there are 1 or 2 real solutions, also print the solutions.

Notes:

1. If $a \neq 0$ and there are real solutions to the equation, you can get these solutions using the following formula:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The number of solutions depends on whether $(b^2 - 4ac)$ is positive, zero, or negative.

2. In order to calculate the square root of a number (of type `double`), you should call the `sqrt` function, located in the `cmath` library.

Your program should interact with the user **exactly** as it shows in the following example:

```
Please enter value of a: 1
Please enter value of b: 4
Please enter value of c: 4
This equation has a single real solution x=-2.0
```

Question 4

Define the following constants:

```
const int FLOOR_ROUND = 1;
const int CEILING_ROUND = 2;
const int ROUND = 3;
```

Write a program that asks the user to enter a Real number, then it asks the user to enter the method by which they want to round that number (floor, ceiling or to the nearest integer). The program will then print the rounded result.

Your program should interact with the user **exactly** as it shows in the following example:

```
Please enter a Real number:
4.78
Choose your rounding method:
1. Floor round
2. Ceiling round
3. Round to the nearest whole number
2
5
```

Implementation requirements:

1. Use a `switch` statement.
2. You are **not** allowed to include and use the `math` library.